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AlcoProtect™

A GUT–LIVER–BRAIN AXIS PROBIOTIC & POSTBIOTIC PLATFORM

Structured for industry partners, behavioral health innovators, liver health brands, and clinical collaborators.

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Technology Overview

AlcoProtect™ is a proprietary, human-origin probiotic and postbiotic platform developed to support alcohol metabolism and modulate gut–liver–brain axis pathways implicated in alcohol-related health burden and addiction cycles.

The strain has been:

- Fully purified and phenotypically characterized
- Whole genome sequenced for identity and safety validation
- WholeScreened to be free of major pathogenicity and transferable antibiotic resistance genes genome sequenced for identity and safety validation
- Deposited in ATCC for traceability and transparency
- Scaled under GMP-compliant manufacturing conditions

Preclinical data demonstrate:

- Enhanced alcohol clearance pathways in hepatic cell models
- Reduction of alcohol-induced cellular stress markers
- Modulation of neurotransmitter-associated pathways linked to addiction behaviors

AlcoProtect™ represents a systems-level microbiome approach targeting alcohol metabolism, liver resilience, and neurochemical balance.

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Public Health Need

Alcohol misuse remains a significant global health challenge, contributing to:

- Fatty liver disease and progressive liver damage
- Neurological impairment
- Behavioral dependence cycles
- Increased intestinal permeability ("leaky gut")
- Systemic inflammation

Current interventions focus primarily on behavioral modification or pharmacologic therapy. There is increasing demand for adjunct microbiome-based strategies that:

- Support metabolic clearance
- Promote gut barrier health
- Address gut–brain signaling pathways

Innovation Rationale

AlcoProtect™ was developed to simultaneously target:

- Gut microbiome modulation
- Hepatic alcohol metabolism pathways
- Neurotransmitter balance within the gut–brain axis

This multi-axis approach provides a differentiated, biologically rational platform that supports alcohol-related health management beyond single-mechanism solutions.

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STRAIN CHARACTERIZATION

Human-Origin Strain Development

AlcoProtect™ is derived from a human-origin microbial isolate selected based on:

- Functional screening in alcohol metabolism models
- Compatibility with intestinal and hepatic systems
- Favorable probiotic characteristics

The strain is:

- Gram-positive
- Catalase-negative
- Stable under fermentation conditions

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Genomic Sequencing & Safety Profiling

Whole genome sequencing confirms:

- Absence of significant virulence-associated genes
- No major transferable antibiotic resistance determinants
- Stable chromosomal structure
- Genetic consistency across production batches

Safety evaluations align with:

- GRAS framework expectations
- EFSA QPS considerations
- International probiotic safety guidelines

Antibiotic Resistance Screening

Comprehensive genomic and phenotypic screening confirms:

- No clinically significant resistance determinants
- No high-risk mobile genetic elements

ATCC Deposition

ATCC deposition provides:

- Scientific traceability
- Independent verification
- IP integrity
- Regulatory transparency

This strengthens sponsor confidence and commercialization readiness.

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MECHANISTIC EVIDENCE

LIVER CELL MODEL FINDINGS

Controlled in vitro hepatic cell studies demonstrate:

- Enhanced alcohol clearance-associated enzymatic pathways
- Reduction in alcohol-induced oxidative stress markers
- Improved cellular resilience under ethanol exposure conditions

These data support potential modulation of hepatic metabolic processes.

GUT–LIVER–BRAIN AXIS INTERACTIONS

Emerging research demonstrates bidirectional communication between:

- Gut microbiota
- Liver metabolism
- Central nervous system signaling

Preliminary investigations suggest AlcoProtect™ may influence:

- Neurotransmitter-associated pathways involved in behavioral regulation
- Signaling molecules linked to craving cycles
- Inflammatory mediators affecting gut–brain communication

These findings support further exploration of addiction-modulating pathways.

POSTBIOTIC ACTIVITY

Cell-derived and heat-treated fractions demonstrate:

- Retained bioactivity in cellular models
- Potential metabolic signaling influence independent of live-cell colonization
- Improved formulation flexibility and stability

This allows development of:

- Live probiotic formats
- Postbiotic-enhanced formulations
- Hybrid gut–brain support products

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DEVELOPMENT & REGULATORY

GMP Manufacturing

AlcoProtect™ is currently:

- Manufactured in GMP-certified facilities
- Produced under controlled fermentation protocols
- Subject to identity, purity, and potency testing
- Evaluated for stability and shelf-life

Quality control includes:

- Batch-level genomic identity verification
- Microbial purity testing
- Contaminant screening
- Viability and potency validation

GRAS Preparation

GRAS pathway preparation includes:

- Genomic safety documentation
- Toxicological data review
- Manufacturing validation
- Expert panel assessment

This supports safe positioning for human consumption.

Clinical Development

Planned clinical studies aim to assess:

- Alcohol metabolism biomarkers
- Liver function markers (e.g., ALT, AST, GGT)
- Inflammatory markers
- Gut barrier integrity
- Behavioral and craving-related endpoints
- Microbiome compositional shifts

Randomized, placebo-controlled designs are under development.

Potential Regulatory Positioning

Possible categories include:

- Dietary supplement
- Liver health formulation
- Behavioral health adjunct
- Preventive wellness support

Claims strategy will align with regulatory frameworks and substantiation data.

COMMERCIAL OPPORTUNITIES

TARGET APPLICATIONS

AlcoProtect™ may support:

- Alcohol metabolism support supplements
- Liver health formulations
- Behavioral wellness adjunct programs
- Corporate wellness initiatives
- Preventive alcohol-use support products

PRODUCT FORMATS

Potential formats include:

- Capsules
- Synbiotic blends
- Functional beverages
- Postbiotic-enhanced sachets

PARTNERSHIP MODELS

MusB Research offers flexible collaboration options:

- Licensing agreements
- Co-development partnerships
- Joint clinical validation programs
- Exclusive distribution arrangements
- Strategic behavioral health collaborations

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STRATEGIC DIFFERENTIATION

AlcoProtect™ offers:

- Multi-axis gut–liver–brain targeting
- Human-origin, genomically transparent strain
- Mechanistic preclinical validation
- GMP-compliant scale-up
- Regulatory-ready development pathway
- Strong differentiation in behavioral and metabolic wellness categories

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AlcoProtect™

AlcoProtect™ represents a novel microbiome-based strategy addressing alcohol-related health burden through integrated gut, liver, and neurochemical pathways. By combining rigorous genomic characterization, mechanistic validation, and regulatory preparation, MusB Research provides a scalable and scientifically defensible platform for innovation in alcohol metabolism and behavioral health support.

As public health systems seek safer, adjunctive, and preventive strategies, AlcoProtect™ offers forward-thinking partners an opportunity to lead in the next generation of microbiome-driven wellness solutions.

MusB Research welcomes strategic collaborators ready to advance evidence-based innovation in gut–liver–brain health.